

■ SUB – ELECTRONICS COMPONENTS AND DEVICES [Join jharkhand polytechnic WhatsApp group -9234415244](#)

■ UNIT – 1: Basic of Electronic Components

Q1. Explain the role of passive components (resistors, inductors and capacitors) and write its SI units and symbols.

Q2. Classify resistors based on their construction and application.

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Q3. Write the difference between inductors and capacitors with applications.

Q4. Explain the usage of electronic instruments – multimeter and LCR meter.

Q5. Explain BJT with its functions and symbols.

Q6. Explain MOSFET. Write the construction and working principle.

Q7. Explain covalent bonding in semiconductors with neat diagram. How intrinsic and extrinsic semiconductor works?

Q8. Differentiate between active and passive components with example and give its function of a multimeter and LCR meter in electronics.

Q9. Define PN-junction diode. Explain the formation of PN junction.

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Q10. Compare conductors, insulators and semiconductors with examples.

Q11. What is the concept of simple electronic circuit? List the components and explain their advantages.

Q12. Explain the different types of special purpose diodes with neat diagram.

Q13. Explain SCR, DIAC and TRIAC. Write their functions, symbols and applications.

Q14. Explain the construction, operation and characteristics of TRIAC.

UNIT – 2: BASICS OF SEMICONDUCTOR DEVICES

1. Explain depletion region and write the formation of depletion region.
2. Explain barrier potential or junction potential.
3. Explain forward biasing and reverse biasing of diode and characteristics.
4. Write reverse V-I characteristics and applications.
5. Explain Zener diode working principle, construction and write its features.
6. Difference between avalanche and Zener breakdown.
7. Explain the terms triggering and commutation in SCR.
8. Explain SCR and write its construction and working.
9. Write short notes on: (a) Peak Inverse Voltage (PIV) (b) Form factor and Peak factor (c) Rectification efficiency (d) Ripple factor (e) Transformer Utilisation Factor.
10. What is rectifier and its need? Types of rectifier.

UNIT – 3: OPTO ELECTRONIC DEVICES

11. Compare between half wave and full wave rectifiers.
12. What is voltage regulation? Explain the working principle of Zener diode as a voltage regulator.
13. Explain MOSFET construction and working principle.
14. With diagrams, explain half wave and full wave rectification. Compare their efficiency and ripple factor.
15. With a neat diagram, explain the working of BJT. Describe the functions of the base, collector and emitter.
16. Explain bridge rectifier.

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UNIT – 3: OPTO ELECTRONIC DEVICES

1. Define opto-electronic devices and write its working principle with examples.
2. Explain photo emission, photoconduction and photovoltaic effects with suitable examples.
3. Explain the concept of photoelectric effect and its application in opto-electronic devices.
4. Explain how biasing affects the operation of an LED. What happens when an LED is forward biased and reverse biased?
5. What are the various packages available for LEDs?
6. Explain LDR and write its construction and working principle.
7. What is a photodiode? Describe its construction and working principle.
8. Define phototransistors. Describe its working principle and construction.

Q1. What is the difference between sensors and actuators?

Q2. Define sensors. Classify them into passive and active sensors. Explain their working principle with some examples of each.

Q3. Define actuators. Classify actuators into passive and active types. Explain their working principle with one example each.